



Cross-regional manpower scheduling and routing problem with stochastic service times in home health care

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ABSTRACT

At present, home health care services supply is still insufficient and unbalanced in China, and the elderly nursing manpower is particularly scarce. Therefore, it is important to reasonably arrange the existing nursing personnel to make full use of the service capabilities. This paper proposes a mixed integer linear programming (MILP) model for a cross-regional scheduling and routing problem with stochastic service times in the one-day planning horizon. A range of possible realistic constraints are considered, including patient time windows, working time regulations, task importance, and skill matching. Due to the complexity of the problem, a hybrid Simulated Annealing algorithm is introduced to solve the model. Monte Carlo simulation method is used for the uncertainty cost evaluation. The numerical example tests verify the effectiveness of the algorithm proposed. This paper analyzes the advantages and disadvantages of cross-regional scheduling and routing by comparing the empirical experiment results of the cross-regional and independent scheduling of two regions. It is found that in the case of this paper, cross-regional scheduling is generally beneficial to reducing the cost of both care centers and improving the efficiency of resource utilization.

1. Introduction

The aging of the population is a common social phenomenon worldwide, which brings about a series of health management problems (He et al., 2016). Like many countries, China faces a serious aging problem. According to the relevant statistics published by the seventh national population census, the population aged 65 and over accounts for 13.5% of the total population by the end of 2020 in China. As age increases, numerous underlying physiological changes occur, and the risk of chronic disease rises (WHO, 2015). As the aging problem worsens, the medical resources of public hospitals are still scarce and cannot meet the enormous demands. In this context, Home Health Care (HHC) has become an efficient way to provide medical care for the elderly and thus constitutes an important part of the community endowment.

HHC refers to the medical services being provided in the patients' homes rather than in public hospitals. Caregivers can meet the patients' needs by bringing necessary equipment or drugs to the patients' homes and providing care therein (Restrepo et al., 2020). Conditions requiring HHC services most frequently include diabetes, heart failure, hypertension, osteoarthritis, chronic ulcer of the skin, and so on (Cappanera et al., 2018). Due to the outbreak of COVID-19, the demand for HHC increases as many people are asked to stay at home while they still need

health and medical services, especially for those elderly people who are hard to get around or susceptible to the virus (Nikzad et al., 2021).

In door-to-door services of HHC, caregivers are scheduled and routed to perform various services at the patients' homes. As this process requires a combination of vehicle routing and scheduling approaches, optimization problems are complex and of high interest to stakeholders such as researchers, practitioners, and policy-makers (Fikar & Hirsch, 2017). Due to the limited resources and realistic constraints of HHC, the process of allocating and scheduling the nursing manpower becomes more complicated and difficult compared with similar Vehicle Routing Problem with Time Window (VRPTW). Many researchers have studied this kind of problem as the Home Health Care Routing and Scheduling Problem (HHCRSP) under different national and regulatory settings. For instance, in context of china, Li et al. (2022) presented a multi-objective HHCRSP in a fuzzy environment that considers each customer has different priorities and the uncertainty of travel and service time. However, few researches take the cross-regional collaborative scheduling and routing problem into account.

The distribution of home medical care resources is unbalanced and inadequate in China (Zheng et al., 2020). The HHC industry is restricted by insufficient nursing manpower and scarce service items (Hao et al.,

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2020). Therefore, it is necessary to make full use of the existing nursing manpower to improve the operation efficiency of the HHC institutions. Due to insufficient supply of elderly nursing personnel and unreasonable talent structure, some HHC institutions can provide limited types and quantities of services. In this condition, cross-centers or cross-regional cooperation can achieve the global optimal and improve the efficiency of resource utilization. This paper aims at a cross-regional scheduling and routing problem for nursing manpower with stochastic service times. The main difference between cross-regional and independent scheduling is whether there is a cooperative relationship between different institutions. In researches related to independent scheduling, only one care center is considered, whether the scheduling is in a one-depot way or a multi-depot way. The scheduling objective only needs to measure the benefits and costs of one care center. In contrast, two or more care centers should be considered in cross-regional scheduling. Each care center operates based on its own interests, so service pricing is an important factor when one of the care centers provides services to another. To describe the cross-regional scheduling, a mixed integer linear programming model is proposed with the objective of minimizing the total dispatching cost. A range of possible realistic constraints are considered, including patient time windows, working time regulations, task importance, and skill matching. The Monte Carlo simulation method and a hybrid Simulated Annealing algorithm is optimized to solve the proposed model. The instances for empirical experiments are generated based on the real-world data of two HHC enterprises in Minhang District, Shanghai. By comparing the results of cross-regional and independent scheduling of caregivers for two regions, this paper proves the superiority of cross-regional scheduling.

The main academic contribution of this paper is threefold. Firstly, this paper initially considers the cross-regional collaborative scheduling of the nursing manpower from a global perspective, which is a less considered problem setting with practical significance. Secondly, few studies have considered the impact of the benefit of care center, and the consideration of indicators related to patient and caregiver satisfaction is not comprehensive enough. The scheduling objectives of this study fully considers factors related to care center costs, patient satisfaction and caregiver satisfaction, and introduces service benefits and cross-regional scheduling benefits into care center costs, which provides new insights into the goal setting of related research. Finally, in the of solution methods, this study proposes a hybrid Simulated Annealing algorithm framework, and introduces the solution acceptance criteria of Simulated Annealing algorithm into the Monte Carlo simulation, which enriches the relevant algorithms in the field of HHCRSP. This paper has some practical contributions. Firstly, for profit-oriented HHC enterprises, the results of this study can be directly applied to caregiver scheduling considering inter-enterprise collaboration, confirming the cross-regional staff scheduling plan, so that the service plans of HHC enterprises can be carried out in an orderly manner. Secondly, the scheduling method in this study is conducive to reducing the operating cost of HHC enterprises, improving patient satisfaction, and reducing the workload of nursing personnel.

The rest of this paper is organized as follows. Section 2 reviews the abundant literature related to our problem in the HHC field. Section 3 is the proposed model, including problem description, notations and mathematical formulation. In Section 4, the solution methodology is conducted, followed by a series of numerical experiments. In Section 5, empirical experiments are carried out based on actual data, followed by a series of sensitivity analyses. Afterwards, concluding and future remarks are presented in Section 6.

2. Literature review

As is illustrated by Redjem and Marcon (2016), operations management in HHC can be classified into two levels. The first level addresses strategic decisions, such as the resources sizing, the splitting or the districting of the covered space and the assignment of resources to the

districts. The second level addresses the operational decisions, such as the visits assignment to caregivers and the problem of caregivers routing. According to the planning horizon, the operational decisions can be divided into the daily scheduling and routing problem, and the periodic staffing and scheduling problem. In the remainder of this section, we will present studies related to daily scheduling and routing problem, which is more relevant to our research. For a wider range of topics in the HHC operations management domain, the interested reader can see Cappanera and Scutellà (2022), Cissé et al. (2017), Di Mascolo et al. (2021), Emiliano et al. (2017), Fikar and Hirsch (2017), Grieco et al. (2020), Marcon et al. (2016), and Wan et al. (2020).

In general, the related literature can be classified according to whether stochastic problems are considered. Thus, we first review papers dealing with deterministic problems. Firstly, travel costs are one of the most important factors in the HHCRSP problem. Fathollahi-Fard et al. (2020, 2018a, 2018b) and Liu et al. (2013) considered minimizing travel costs in their models. Liu et al. (2013) focused on the delivery of drugs and medical devices, and proposed a Genetic Algorithm (GA) and a TS method to solve the mixed-integer programming models. Fathollahi-Fard et al. (2018a) firstly attempted to consider the issue of green HHC. The authors developed a bi-objective optimization model in which the green emissions and environmental pollution of HHC organization are considered. A number of modification and hybrid versions of metaheuristics as well as four fast heuristics are constructed to solve the model. Fathollahi-Fard et al. (2018b) developed a single objective optimization MILP model and employed a solution approach based on Lagrange relaxation theory. Based on Fathollahi-Fard et al. (2018b), the authors proposed three new heuristics and a hybrid constructive metaheuristic in Fathollahi-Fard et al. (2020). Different from the above studies, Akjiratikar et al. (2007) and Riazi et al. (2014, 2018) considered minimizing travel distance in their papers. Specifically, Akjiratikar et al. (2007) proposed a Particle Swarm Optimization (PSO) algorithm to solve the scheduling problem of home care workers in order to minimize the distance traveled. Riazi et al. (2014) solved the HHCRSP by a distributed Gossip algorithm. Based on the algorithm proposed in Riazi et al. (2014, 2018) integrated the Gossip algorithm with a local solver based on the combining column generation framework to solve larger problem instances. Erdem and Koç (2019) minimized the total travel time by considering heterogeneous fleets, time windows, preferences, competencies, connected activities, the range of electric vehicles, charging status, and charging strategies. A hybrid metaheuristic combining genetic algorithm and a variable neighborhood descent is developed to solve the problem.

Cost reduction is one of the main objectives in scheduling and routing for HHC organizations. In addition to travel costs, personnel-related costs are also important optimization goals. For example, Decerle et al. (2016) aimed to minimize the cost related to transportation and working hours, and develop a two-phase matheuristic in order to distinguish different kinds of nursing personnel. Fikar and Hirsch (2015) defined the problem as an extended many-to-many multi-trip DARP. To solve the problem, they introduced a metaheuristic with two stages, identifying potential walking routes and optimizing the transport system. Liu et al. (2021) considered the factors of synchronized visits, lunch breaks and flexible departure modes of caregivers. The authors aimed to minimize the wages of caregivers and the costs of vehicles. Then, four hybrid metaheuristics are developed to solve the model. Samadpour et al. (2022) developed a MILP model to investigate how to plan and schedule health workers' workdays, allocate nurses and doctors appropriately to meet patient needs, and make accurate decisions to minimize costs.

Some researchers consider fixed costs in their studies, such as Fathollahi-Fard et al. (2022), Liu, Tao et al. (2019) and Nasir and Kuo (2020). Liu, Tao et al. (2019) considered fixed costs of vehicles and synchronized-services constraints in their model. To save the computational time, an efficient Adaptive Large Neighborhood Search heuristic

is proposed to solve the problem. Nasir and Kuo (2020) considered fixed costs of both vehicles and nurses. The authors solved the challenges of creating simultaneous schedules and route plans for a set of HHC staff and Home Delivery Vehicles (HDVs) under the requirements of synchronization between HHC staff and HDVs visits, multiple visits to patients, multiple routes of HDVs and pickup/ delivery visits related precedence for HDVs. A Hybrid Genetic Algorithm (HGA) was proposed to suggest HHC planning decisions. Fathollahi-Fard et al. (2022) firstly considered outsourcing for patients' demand to receive direct services from a hospital as a real assumption. This work developed a bi-level programming model as a static Stackelberg game between nurses and patients within the framework of HHCSC. Meta-heuristics and hybrid ones were utilized to get feasible solutions.

Patients' satisfaction or caregivers' satisfaction becomes part of the optimization goals in some studies. Braekers et al. (2016) and Li et al. (2021) took the patients' satisfaction into account. Braekers et al. (2016) shed light on the trade-off relationship between the two objectives between costs and client inconvenience. They proposed a metaheuristic that embeds a large neighborhood search heuristic in a multi-directional local search framework to solve the problem. Li et al. (2021) set maximizing the total benefit of patients' preference satisfaction as one of the objective functions. An outer-approximation method was adopted to obtain the global ϵ -optimal solutions for the small-scale problem when a hybrid genetic algorithm was developed to solve the large problem. Different from Braekers et al. (2016) and Li et al. (2021), Trautsumwieser and Hirsch (2011) considered both patients' and caregivers' satisfaction. A metaheuristic method based on the Variable Neighborhood Search was conducted to solve small problem instances. For the patients, whether the caregivers provide services on time will have an impact on the patients' satisfaction. Bertels and Fahle (2006), Decerle et al. (2019), Hiermann et al. (2015) and Manavizadeh et al. (2020) considered penalties for unvisited patients or uncovered tasks in their objective functions in different ways. To solve the proposed problem, Bertels and Fahle (2006) improved and combined heuristics like SA and TS; Decerle et al. (2019) developed an original hybrid algorithm combining memetic and ant colony optimization algorithm; Hiermann et al. (2015) constructed four metaheuristics: variable neighborhood search, a memetic algorithm, scatter search and a simulated annealing hyper-heuristic; Manavizadeh et al. (2020) utilized the Simulated Annealing (SA) algorithm.

In most researches, the scheduling problem is based on the assumption that all patients' demands should be met. However, HHC nursing staff are in short supply in reality. Only a few studies suppose that there exist unvisited patients, and consider the objective related to the unvisited patients in the scheduling model. Dohn et al. (2009) considered the manpower allocation problem with job-teaming constraints and a limited number of teams (m-MAPTWC) in their model. A column generation in a branch-and-price framework is presented to solve the model. Erdem and Bulkan (2017) introduced a two-stage solution approach to solve the large-scale problem. In the first stage, a cluster-assign algorithm is employed to obtain feasible clusters. In the second stage, using these clusters, route construction heuristics are utilized to create schedules and routes. Rasmussen et al. (2012) modeled the problem as a set partitioning problem with side constraints and developed an exact branch-and-price solution algorithm. Liu et al. (2017) addressed a scheduling and routing problem with the consideration of lunch break requirements and solved the problem based on the proposed branch-and-price (B&P) algorithm.

Moreover, few studies have considered service revenues in the scheduling problem. Manerba and Mansini (2016) aimed to maximizing the benefit gained by performing the services considering potential incompatibilities among pairs of services for a single patient. Lin et al. (2021) studied the scheduling problem in HHC from a centralized perspective of service-sharing platforms. The authors maximize the difference between the caregiver's received income and spent costs, which represents the average caregiver's profit. To solve the problem, a

branch-and-price approach was designed to match demand with supply using different matching strategies. However, both Lin et al. (2021) and Manerba and Mansini (2016) do not consider the stochastic factors in their studies.

Subsequently, we review papers dealing with stochastic problems. Table 1 summarizes the stochastic factors and methods in these studies. Shi et al. (2017) was the first to incorporate the vehicle scheduling of visiting patients and the fuzzy demand of each patient. Stochastic simulation and Hybrid Genetic Algorithm (HGA) are integrated to solve the problem. Through robust optimization method, Cappanera and Scutellà (2022) solved the demand uncertainty, people-oriented consistency and arrival time consistency in home care scheduling problem. Most researches consider stochastic travel and service times in their studies. El-Amraoui et al. (2022) adopted robust optimization approaches to model and deal with the uncertainty of travel time, the dynamics of the system and the variability of patient data. Yuan et al. (2015) addressed a HHC scheduling and routing problem with stochastic service times and skill requirements. To solve the problem, it is equivalently transformed into a master problem and a pricing sub-problem. The column generation procedure is embedded within the B&P framework to obtain feasible solutions. In Yuan et al. (2015), travel times between two customers is deterministic, and service times to a patient is stochastic due to their varying health conditions. Similar to Yang et al. (2018), Yuan et al. (2015) and Zhan et al. (2021) assumed that service times are stochastic. Yang et al. (2018) studied the scheduling problem in dense communities, where traveling distance is shortened and waiting cost is possibly higher. A chance constrained programming model is built to formulate the problem, and a heuristic based on the Best-Worst Ant Colony Optimization is developed to solve it. Zhan et al. (2021) solved the problem of stochastic service times by formulating a stochastic programming with recourse. The authors develop an L-shaped method to efficiently solve the problem, and propose an easy-to-implement heuristic algorithm that allows for effectively solving large instances. Some researchers considered the stochastic travel and service times at the same time. Liu, Yuan et al. (2019) introduced chance constraints to ensure the on-time service probability for the customers. A branch-and-price (B&P) algorithm and a discrete approximation method are combined to solve the problem. Hashemi Doulabi et al. (2020) and Shi et al. (2018) used the method of stochastic programming with recourse to consider stochastic travel and service times. Shi et al. (2018) aimed to minimize the total transportation cost and the recourse cost. A metaheuristic Simulated Annealing algorithm was developed to solve the problem and compared with Hybrid GA (HGA), Bat Algorithm (BA) and Firefly Algorithm (FA). Hashemi Doulabi et al. (2020) studied vehicle routing problems with synchronized visits (VRPS) and introduced an operating room scheduling problem with stochastic durations as a novel application of VRPS. L-shaped algorithm and its branch-and-cut implementation was utilized to solve the proposed two-stage stochastic integer programming model. Bazirha et al. (2022) investigated HHCRSP with multiple hard/fixed time windows and stochastic travel and service times. Monte Carlo simulation embedded in GA was used to solve the two-stage stochastic programming with recourse. Shahnejat-Bushehri et al. (2021) and Shi et al. (2019) proposed robust optimization models to consider stochastic travel and service times. Shi et al. (2019) aimed to minimize the total cost among the worst cases. Gurobi Solver, SA, VNS, and TS methods are used to test the validity of the proposed model. Shahnejat-Bushehri et al. (2021) proposed a robust optimization model with the objective of minimizing the total costs. Due to the complexity of this problem, Simulated Annealing, Genetic Algorithm, and Memetic Algorithm were proposed to solve the developed model.

By reviewing the papers that addressed the HHCRSP, it can be concluded that few researchers pay attention to the cross-regional scheduling problem. In some nursing centers, the nursing manpower is insufficient or lacks skill qualification, so the center cannot provide all the services required by the contracted patients. In this case,

Table 1
Stochastic factors and methods related to the HHCRSP in the literature.

Paper	Uncertainty			Method		
	DD	TT	ST	CCP	SPR	RO
Bazirha et al. (2022)		✓	✓		✓	
Cappanera and Scutellà (2022)	✓					✓
El-Amraoui et al. (2022)		✓				✓
Hashemi Doulabi et al. (2020)		✓	✓		✓	
Liu, Yuan et al. (2019)		✓	✓	✓		
Shahnejat-Bushehri et al. (2021)		✓	✓			✓
Shi et al. (2017)	✓			✓		
Shi et al. (2018)		✓	✓		✓	
Shi et al. (2019)		✓	✓			✓
Yang et al. (2018)			✓	✓		
Yuan et al. (2015)			✓		✓	
Zhan et al. (2021)			✓		✓	

cross-regional cooperation can supplement the lack of service capacity and improve patient satisfaction. Thus, this paper proposes the cross-regional scheduling problem and the corresponding model. A hybrid Simulated Annealing algorithm is constructed to solve the proposed problem. Moreover, this paper compares the scheduling results of cross-regional and independent scheduling to analyze the advantages and disadvantages of cross-regional scheduling.

3. Problem description

In this section, a brief description of the cross-regional nursing manpower scheduling problem is proposed. In order to simplify the problem, two nursing centers conducting cross-regional scheduling are considered. It is assumed that there are two regions, which are Region A and Region B, respectively. Each region has one care center, and patients in each region will sign up with the appropriate care center. The required assumptions are defined as follows:

- (1) Care centers in region A and Region B can serve patients in the other region through signing cooperation agreements and paying for cross-regional services.
- (2) In a one-day planning horizon, all caregivers depart from their care centers, serve each patient in turn, and return to the care center.
- (3) Each patient can request nursing service once at most.
- (4) As same as Dohn et al. (2009), Erdem and Bulkan (2017) and Rasmussen et al. (2012), some patients cannot receive any service in the planning horizon due to the limited number of caregivers.
- (5) The services are divided into normal services and vital services. The vital services must be completed within the time window. The normal services are allowed to be down in violation of the time windows or not to be down.
- (6) The caregivers have working time regulations and working overtime will be paid with extra wages.
- (7) In order to avoid work exhaustion and ensure service quality, there is a limited number of services assigned to a caregiver.
- (8) The caregivers can serve patients who require services with equal or lower skill qualifications, and the penalty costs will be imposed for overqualification.
- (9) Each patient has a service time window. If the caregiver arrives earlier than the left time window, he or she needs to wait until the earliest time window before starting the service; otherwise, the caregiver needs to start the service immediately. If the caregiver arrives later than the right time window, the penalty costs for the time of delay should be calculated.
- (10) The coordinates and time windows of patients, and the travel distance and times between patients are known before scheduling.
- (11) The service duration is uncertain due to the different health conditions of patients.

3.1. Notion

In this section, we define the cross-regional scheduling and routing problem for nursing manpower in home health care. Given a directed graph $G = (V, A)$, where $V = \{0, m_A + m_B + 1\} \cup P$ is the set of nodes, and $A = \{(i, j) \mid i, j \in V, i \neq j\}$ is the set of arcs. $P_A = \{1, 2, \dots, m_A\}$ is the set of patients, in which $P_A = \{1, 2, \dots, m_A\}$ is the set of patients in the region A, and $P_B = \{m_A + 1, m_A + 2, \dots, m_A + m_B\}$ is the set of patients in the region B. P_{imp} is the set of vital services. 0 and $m_A + m_B + 1$ represent nodes of the care center A and the care center B. A is characterized by the travel distance d_{ij} , travel times t_{ij} , and the travel cost c_{ij} . $N = N_A \cup N_B$ is the set of caregivers of both care centers, where N_A is the set of caregivers of the care center A, and N_B is the set of caregivers of the care center B.

In this paper, we use three-dimensional coordinates (x_i, y_i, z_i) to represent the location of patients. d_{ij} is calculated by the formulation (1). Suppose that the travel time t_{ij} is proportional to the travel distance. In the dense Chinese community, caregivers often visit patients by electric bicycles or on foot. Thus, the travel time t_{ij} is calculated by the formulation (2).

$$d_{ij} = |x_i - x_j| + |y_i - y_j| + z_i + z_j \quad (1)$$

$$t_{ij} = (|x_i - x_j| + |y_i - y_j|) / v_e + (z_i + z_j) / v_f \quad (2)$$

Inspired by Yuan et al. (2015), we use dummy qualification demands and dummy qualification capabilities to describe the skill constraints and calculate the overqualification penalties. As well as Yuan et al. (2015), we give an example to illustrate the method. Suppose that there are ten customers, in which six customers require the services with the skill level 1 and the others with the skill level 2. Also, there are two caregivers who belong to the two skill levels. Each patient who requires the service with the skill level 1 has a dummy demand of 10 units, and each caregiver at the skill level 1 has a dummy capacity of 50 units. Each patient who requires the service with the skill level 2 has a dummy demand of 60 units, and each caregiver at the skill level 2 has a dummy capacity of 300 units. Then, the overqualification penalty will be 240 (300–60) units. The notations of parameters and decision variables for the proposed mathematical formulation are summarized as follows:

Parameters:

i, j	Index of nodes
n	Index of caregivers
σ	Set of skill qualification for different types of services ($\sigma = \{1, \dots, s\}$)
σ_i	Skill qualification of the service needed by the patient i
σ_n	Skill qualification owned by the caregiver n
T_a	The maximum number of patients served by a caregiver
a_i, b_i	The earliest and latest time window of the patient i
β	Proportion of the vital services
a_n, b_n	The working time window of the caregiver n
d_{ij}, t_{ij}, c_{ij}	Travel distance, travel times and costs between the nodes i and j
v_e	Travel speed by using electric bicycles
v_f	Travel speed on foot
r_i	The time that the caregiver reaches the patient i
s_i	The time that the caregiver starts to serve the patient i
w_i	The duration that the caregiver waits to serve the patient i
l_i	The duration that the caregiver delays to serve the patient i

d_i	The service duration required by the patient i
r_s	The unit revenue for serving a patient, which is related to the skill qualification required by the service
c_{it}	The unit travel time cost
c_{pt}	The unit penalty for violating the patients' time windows
o_n	Overtime of the caregiver n
c_{sf}	The unit fixed cost for hiring a caregiver with the skill qualification s
c_{sv}	The unit overtime cost for a caregiver with the skill qualification s
q_{si}	Dummy demands of the patient i who requires the service with the skill qualification s
Q_{sn}	Dummy capacities of the caregiver n with the skill qualification s
c_d	The unit penalty for overqualification
c_w	The unit penalty for waiting times
c_{cus}	The unit penalty for unvisited patients
c_{nur}	The unit penalty for overtime
r_{cross}	The unit revenue/cost for providing/receiving cross-regional services

Decision variables:

x_{ijn}	1, if the caregiver n serves the patient j after serving the patient i ; 0, otherwise
y_i	1, if the patient i receives the service; 0, otherwise
y_{in}	1, if the caregiver n serves the patient i ; 0, otherwise
z_i	1, if the patient i receives the cross-regional service; 0, otherwise

3.2. Mathematical model

This study aims to minimize the total scheduling costs of the two care centers. The proposed mathematical model for the cross-regional scheduling and routing problem is presented as follows:

$$\begin{aligned}
 \min f_1 = & \sum_{n \in N} c_{sf} \delta_n + \sum_{i,j \in V} \sum_{n \in N} c_{ij} x_{ijn} + \sum_{n \in N} c_{sv} o_n - \sum_{i \in P} \sum_{n \in N} r_{sn} y_{in} \\
 [htbp] & + \sum_{i \in P} \sum_{n \in N} c_{cus} (1 - y_{in}) + \sum_{i \in P} \sum_{n \in N} c_{pt} l_i y_{in} + (\max c_{nur} o_n) \\
 & + \sum_{i \in P} \sum_{n \in N: \delta_i < \delta_n} c_d y_{in} + \sum_{i \in P} \sum_{n \in N} c_w w_i y_{in}
 \end{aligned} \quad (3)$$

The first to four terms of objective (3) refer to minimizing the operational cost, which are fixed costs for hiring caregivers, travel costs, overtime costs, and revenue for serving patients, respectively. The five and six terms aim to minimize the patient dissatisfaction, including the penalty for unvisited patients and the penalty for violating the patients' time windows. Finally, the seven to nine terms minimize the caregiver dissatisfaction, including the penalty for working overtime, the penalty for overqualification, and the penalty for waiting times. Since the cross-regional cost is paid from one care center to the other, the cross-regional revenue equals zero when calculating the total cost of both care centers.

In the cross-regional routing and scheduling problem, the scheduling cost of the individual care center is also important. If the scheduling cost in the cross-regional mode is higher than in the independent mode, it will be difficult to motivate the individual care center to join in the cross-regional scheduling. Thus, this paper also calculates the total costs of each care center based on the formulas (4) and (5). f_{1A} and f_{1B} are calculated in the same way as f_1 ; however, only patients and caregivers in the corresponding region will be calculated respectively.

$$\min f_{2A} = f_{1A} + \sum_{i \in P_A} \sum_{n \in N_B} r_{cross} z_i y_{in} - \sum_{i \in P_B} \sum_{n \in N_A} r_{cross} z_i y_{in} \quad (4)$$

$$\min f_{2B} = f_{1B} + \sum_{i \in P_B} \sum_{n \in N_A} r_{cross} z_i y_{in} - \sum_{i \in P_A} \sum_{n \in N_B} r_{cross} z_i y_{in} \quad (5)$$

s.t.

$$\sum_{j \in P} x_{0jn} = 1 \quad \forall n \in N_A \quad (6)$$

$$\sum_{j \in P} x_{j0n} = 1 \quad \forall n \in N_A \quad (7)$$

$$\sum_{j \in P} x_{(m_A+m_B+1)jn} = 1 \quad \forall n \in N_B \quad (8)$$

$$\sum_{j \in P} x_{j(m_A+m_B+1)n} = 1 \quad \forall n \in N_B \quad (9)$$

$$\sum_{i \in V} \sum_{j \in P} x_{ijn} - \sum_{j \in P} \sum_{i \in V} x_{jin} = 0 \quad \forall n \in N \quad (10)$$

$$s_i + d_i + t_{ij} \leq s_j + M(1 - x_{ijn}) \quad \forall i, j \in P, n \in N \quad (11)$$

$$s_i \geq y_i a_i \quad \forall i \in P \setminus P_{imp} \quad (12)$$

$$a_i \leq s_i \leq b_i \quad \forall i \in P_{imp} \quad (13)$$

$$\sum_{i,j \in S, i \neq j} x_{ijn} \leq |S| - 1 \quad \forall n \in N, S \in V, |S| \geq 2 \quad (14)$$

$$x_{ijn} \delta_j \leq \delta_n \quad \forall j \in P, n \in N \quad (15)$$

$$\sum_{i \in P} y_{in} q_{si} \leq Q_{sn} \quad \forall n \in N \quad (16)$$

$$\sum_{i \in P} y_{in} \leq T_a \quad \forall n \in N \quad (17)$$

$$x_{ijn} = \{0, 1\} \quad \forall i, j \in V, n \in N \quad (18)$$

$$y_{in} = \{0, 1\} \quad \forall i \in P, n \in K \quad (19)$$

$$y_i = \{0, 1\} \quad \forall i \in P \quad (20)$$

$$z_i = \{0, 1\} \quad \forall i \in P \quad (21)$$

Formulas (6)–(21) are constraints. Constraints (6)–(10) ensure that caregivers of both care centers start their tour at their own care center, visit some patients and end at their place of departure. Constraint (11) determines the arrival and starting service time for each patient. Constraints (12) and (13) are time windows constraints. Constraint (14) is the sub-tour elimination constraint. Service qualification match is guaranteed by constraints (15)–(16). Constraint (17) means that each caregiver can only serve a limited number of patients in the planning horizon. Constraints (18)–(21) are binary constraints for decision variables.

3.3. Calculation for uncertain parameters

Since the health condition of patients is hard to determine in advance, the service times are uncertain parameters when making the scheduling plan. Suppose that the service times of each patient obey the normal distribution $\tilde{d}_i \sim N(\mu, \sigma^2)$. Then, the calculation of uncertain parameters related to uncertain service times are as follows:

$$[l] \tilde{w}_l = \max(a_l - \tilde{r}_l, 0) \quad (22)$$

$$\tilde{l}_l = \max(\tilde{r}_l - b_l, 0) \quad (23)$$

$$\widetilde{r_{l+1}} = \tilde{s}_l + \tilde{d}_l + t_{ij} \quad (24)$$

$$\widetilde{s_{l+1}} = \max(a_{i+1}, \tilde{s}_l + \tilde{d}_l + t_{ij}) \quad (25)$$

$$\tilde{o}_n = \max\left(\sum_{i,j \in V, i \neq j} y_{jn} (\tilde{w}_j + \tilde{d}_j + t_{ij}) - (b_n - a_n), 0\right) \quad (26)$$

The formulas (22)–(26) are used to compute the waiting times, the delay times, the reach times, the start times of services, and the overtimes of caregivers, respectively. The uncertain travel times can be calculated in the same way though not considered in this paper.

9	17	15	8	22	Route 1
10	2	1	14	21	Route 2
3	13	7	23	25	Route 3
16	11	20	12	18	Route 4
4	6	5	19	24	Unassigned patients

Fig. 1. Example of a feasible solution.

4. Solution methodology

In this section, we design a hybrid Simulated Annealing algorithm to solve the proposed cross-regional scheduling model. Due to the complexity of the proposed problem, an encoding method with the list table suggested by Shi et al. (2017) is modified to be adopted. In this method, each solution can be converted into a list table. Each feasible route corresponds to a caregiver, and contains the information of the patients served and their sequential order. In the last row of the list table, unassigned patients are stored as a route with no sequence, and participate in the neighborhood search process. According to the above description, a feasible solution with 4 caregivers and 25 patients is shown as Fig. 1.

4.1. Initial section

Due to the complexity of the objectives and constraints of the cross-regional scheduling problem, the algorithm is highly dependent on the performance of the initial solution. Thus, it is necessary to generate the initial solution in a reasonable way. This study constructs the initial solution from the perspective of skill matching and meeting the patients' time window constraints. The skill matching constraint is transformed into the dummy capacity constraints. The specific steps to construct the initial solution are as follows:

- (1) Generate patient information set randomly, including three-dimensional coordinates of patients, service time windows, expected service duration, etc.
- (2) Arrange the patients to be assigned in reverse order according to their skill levels of required services (the patients with the same skill level is sequenced according to the task importance), and arrange the nursing staff to be assigned in reverse order according to their skill levels. We do this step because the skill matching constraint is one of the most important constraints in our research. A number of skill mismatches can lead to a large increase in the objective function.
- (3) Select the first patient in the list of patients to be assigned and the first nurse in the list of nurses to be assigned to determine whether the remaining skill capacity can provide services for the current patient. If so, go to Step 4. If not, go to Step 5. By repeating this step, patients with higher skill demands can be prioritized to be assigned to caregivers with higher skill levels. After scheduling all the patients with higher skill demands, the caregivers with higher skill levels can serve patients with lower skill demands if they still have service capabilities. In the latter case, the penalty for overqualification will be calculated.
- (4) Assign the current patient to the current caregiver, add the current path according to the earliest patient window, and remove the current patient from the list of patients to be assigned.

- (5) Output the current path, add the current patient to the list of the unassigned set, and remove the current patient and current caregiver from the list of patients to be assigned and the list of nursing personnel to be assigned.
- (6) Repeat steps 3 to 5 until all the routes are determined.
- (7) Add the remaining unassigned patients to the set and output the initial solution ω_0 .

The initial solution constructed based on the above steps can satisfy the skill matching and the time window constraints of patients to a certain extent. Also, all the vital services will be assigned in the initial solution.

Algorithm 1: Method for generating an initial solution

Data: patient set P , caregiver set N , dummy skill demands θ_i , dummy skill capabilities θ_n , time windows $[a_i, b_i]$, maximum number of patients served by a caregiver T_a

Result: initial solution ω_0

```

1  $i \leftarrow 1, n \leftarrow 1$ 
2  $Route \leftarrow [], Load \leftarrow 0$ 
3 while  $i \leq \text{numberofpatients}$  and  $n \leq \text{numberofcaregivers}$  do
4   if  $\theta_i \leq \theta_n - Load$  then
5     if  $\text{length}(Route) < T_a$  then
6       if  $\text{isempty}(Route)$  then
7          $Route \leftarrow [Route, i]$ 
8          $Load \leftarrow Load + \theta_i$ 
9          $i \leftarrow i + 1$ 
10      else
11        insert  $i$  into the Route according to the sequence
12        of  $a_i$ 
13         $Load \leftarrow Load + \theta_i$ 
14         $i \leftarrow i + 1$ 
15      end
16    else
17      save Route as  $Route_n$ 
18       $n \leftarrow n + 1$ 
19       $Route \leftarrow []$ 
20       $Load \leftarrow 0$ 
21    end
22  else
23    if  $\text{length}(Route) < T_a$  then
24      insert  $i$  into the unassigned patients set
25       $i \leftarrow i + 1$ 
26    else
27      insert  $i$  into the unassigned patients set
28      save Route as  $Route_n$ 
29       $n \leftarrow n + 1$ 
30       $i \leftarrow i + 1$ 
31       $Route \leftarrow []$ 
32       $Load \leftarrow 0$ 
33    end
34  end
35 insert the left patients into the unassigned patients set

```

4.2. The objective function evaluation

Since this paper assumes that the patients' service times are uncertain, it is difficult to calculate the objective function; thus, the Monte Carlo simulation method is adopted to conduct the solution evaluation. The principle of the Monte Carlo simulation is to obtain approximate solutions for problems with uncertain parameters by generating random information. According to the distribution of patients' service times, the actual service times of each solution with several



Fig. 2. Example of the intra-change operator.



Fig. 3. Example of the inter-change operator.

routes will be simulated for M times, and the objective function will be evaluated respectively. The results of M simulations will be averaged arithmetically to get the expected objective function value. Generally, the sampling times for Monte Carlo simulation should be large enough to ensure the accuracy of cost evaluation, which will sacrifice the computational efficiency. In order to shorten the running time, two designs are adopted in this study.

Firstly, only partial route solutions are selected for simulation. This paper assumes that the patients' service times follow the normal distribution $\tilde{d}_i \sim N(\mu, \sigma^2)$. When σ equals to 0, the stochastic problem will be reduced to a deterministic one. For a newly generated solution, the objective function value $f(\omega')$ for the deterministic problem will be calculated at first. If $f(\omega')$ is better than the current optimal solution, the Monte Carlo simulation process will be carried out to evaluate the cost of the uncertain problem. If $f(\omega')$ is inferior to the current optimal solution, the simulation process will be carried out within a certain probability. Suppose that the cost difference between the current solution and the optimal solution is $\Delta = (f(\omega') - f(\omega))/f(\omega)$, then the probability will be $P = \exp(-100\Delta)$.

Secondly, the cost evaluation process is simplified. For a specific solution, some items of the objective function will not change with stochastic service times. In the process of simulated cost evaluation, only the items related to patients' service durations will be calculated. These items include overtime costs, penalty for violating the patients' time windows, penalty for working overtime, and penalty for waiting times.

4.3. Simulated annealing and neighborhood structure

Simulated Annealing algorithm has been widely used in various combinatorial optimization fields. Also, SA achieves good performance in the HHCRSP field, such as Bertels and Fahle (2006) and Fathollahi-Fard et al. (2018a), Liu et al. (2021), Shahnejat-Bushehri et al. (2021), and Shi et al. (2019). In order to obtain a better solution, this study adopts the following four strategies for neighborhood search with a certain probability:

Intra-change: Two nodes within a route are randomly selected and exchange their positions. As shown in Fig. 2, the nodes 2 and 5 exchange the sequence within the route 1. The neighborhood solution is obtained after the exchange.

Inter-exchange: Two nodes of different routes are randomly selected, and exchange their positions. As shown in Fig. 3, the nodes 2 and 7 exchange their positions with each other.

Algorithm 2: The process of Monte Carlo simulation

Data: information related to patients and caregivers, cost parameters, $maxSimu$

Result: simulated objective function value f_s

```

1  $m \leftarrow 1$ 
2  $f_{best} \leftarrow f_0$ 
3 calculate the deterministic objective function  $f_d$  of the new
  solution
4 if  $f_d \leq f_{best}$  then
5   conduct the simulation process:
6   calculate the costs that are independent on service times  $f_1$ 
7   while  $m \leq maxSimu$  do
8     generate service times following the normal distribution
9     calculate the objective function  $f_m$ 
10     $m \leftarrow m + 1$ 
11  end
12   $f_s = average(f_m) + f_1$ 
13 else
14    $\Delta = (f_d - f_{best})/f_{best}$ 
15    $P = \exp(-1000\Delta)$ 
16   generate  $r = rand()$ 
17   if  $r \leq P$  then
18     conduct the simulation process
19   else
20     continue the neighborhood search
21   end
22 end

```

0-1 move: A node in any route is randomly chosen, and inserted in other routes or in other positions of the same route. As shown in Fig. 4, the node 2 in route 1 is selected, and moved to the unassigned patient set. The node 7 in route 2 is moved forward three positions in the same route.

Reverse: When two nodes of a route are selected, all the nodes between them will realignment in reverse order. As shown in Fig. 5, the nodes between nodes 2 and 8 reverse to get a new neighborhood solution.

The unassigned patient set can be included in the inter-change and 0-1 move operators. Different from the neighborhood search process of scheduling routes, the nodes in the unassigned patients set do not need to be sequenced by order of earliest times windows, and do not obey

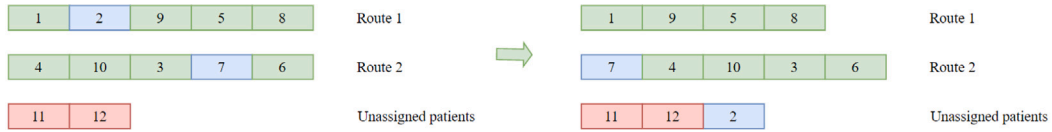


Fig. 4. Example of the 0-1 move operator.



Fig. 5. Example of the reverse operator.



Fig. 6. Distribution of two care centers and nearby communities.

Table 2
Service portfolios and items of the care center A.

Portfolios	Service items				
Assisted nursing	Life care	Medical care			
	–hygiene care	–skin care			
Comprehensive nursing	–diet care	–medicine assistance			
	–cleaning	–excretion care			
Special nursing		–chronic disease care			
	Life care	Medical care	Professional care		
	–hygiene care	–skin care	–pressure ulcer care		
	–diet care	–medicine assistance	–mental care		
	–cleaning	–excretion care	–COPD care		
		–chronic disease care	–rehabilitation care		
			–PICC piping care		
	Life care	Medical care	Professional care	Special care	
	–hygiene care	–skin care	–pressure ulcer care	–postoperative care	
	–diet care	–medicine assistance	–mental care	–care for cancer	
	–cleaning	–excretion care	–COPD care	–care for fracture	
		–chronic disease care	–rehabilitation care	–care for diseases	
			–PICC piping care	–cognitive disorder care	

other constraints either. Also, unassigned patient set has no impact on the objective function evaluation.

5. Empirical experiments

In this section, test instances are generated to analyze the characteristics of cross-regional scheduling based on the real case of two care centers in Minhang District, Shanghai. The basic background of

the care centers is introduced in Section 5.1, and the specific instances information is detailedly described in Section 5.2. The proposed SA algorithm is utilized in Section 5.3 to solve the proposed model. This paper focuses on computing and analyzing the differences between cross-regional and independent scheduling. Afterwards, sensitivity analysis on time windows width and caregivers' skill qualification structure is carried out.

Algorithm 3: Simulated Annealing method

Data: information related to patients and caregivers, cost parameters, $maxInter$, $maxOuter$, T_0 , T_f , R

Result: best solution ω_{best}

```

1  $\omega_{best} \leftarrow \omega_0$ 
2  $f_{best} \leftarrow f_0$ 
3  $T \leftarrow T_0$ 
4  $Inter \leftarrow 0$ ,  $Outer \leftarrow 0$ 
5 while  $T \geq T_f$  do
6   generate neighborhood solution  $\omega_{new}$  by randomly choose
   one operator within the specific probability
7    $f_{current} \leftarrow f_{best}$ 
8   while  $Inter \leq maxInter$  and  $Outer \leq maxOuter$  do
9     if  $f_d \leq f_{best}$  then
10      conduct the simulation process to get  $f_{new}$ 
11       $Inter \leftarrow Inter + 1$ 
12      if  $f_{new} \leq f_{current}$  then
13         $\omega_{current} \leftarrow \omega_{new}$ 
14         $Inter \leftarrow Inter + 1$ 
15        if  $f_{current} \leq f_{best}$  then
16           $\omega_{best} \leftarrow \omega_{current}$ 
17        end
18      else
19        randomly generate  $a \in [0, 1]$ 
20        if  $a \leq \exp(\frac{-(f_{new} - f_{current})}{T})$  then
21           $Inter \leftarrow Inter + 1$ 
22           $\omega_{current} \leftarrow \omega_{new}$ 
23        end
24      end
25    else
26      if  $r \leq P$  then
27        conduct the simulation process to get  $f_{new}$ 
28         $Inter \leftarrow Inter + 1$ 
29      else
30         $Inter \leftarrow Inter + 1$ 
31      end
32    end
33     $Outer \leftarrow Outer + 1$ 
34  end
35   $T \leftarrow R * T$ 
36 end

```

5.1. Case background

There are two care centers A and B in Minhang District of Shanghai. The care center A is a member of a Healthcare Group. As a one-stop medical station, the care center A can provide three service portfolios to the elderly in nearby communities. As shown in Table 2, there are three levels of skill qualification needed to provide different service portfolios. The care center B is one station of a chain health care enterprise with providing HHC services. Compared with the care center A, the care center B is smaller in size, equipped with fewer and less skilled caregivers. The locations of the two care centers and nearby communities are shown in Fig. 6.

5.2. Experiment description

Since cross-regional scheduling is a new problem, where assumptions and constraints are unique, there is no standard benchmark for this version of the HHCRSP. To generate instances to conduct experiments, Solomon's VRTPW benchmark (Solomon, 1987) is adapted according to our proposed problem. There are three types of instances

Table 3

Number of patients and caregivers in each instance.

Size	Region	Number of caregivers				Number of patients			
		$\delta_n = 1$	$\delta_n = 2$	$\delta_n = 3$	Total	$s = 1$	$s = 2$	$s = 3$	Total
50	A	2	2	1	5	12	12	6	30
	B	2	1	0	3	8	8	4	20
100	A	4	4	2	10	24	24	12	60
	B	3	2	1	6	16	16	8	40
200	A	8	8	4	20	48	48	24	120
	B	6	4	2	12	32	32	16	80

Table 4

The set of parameters for objective function evaluation.

Parameter	Value	Unit
v_e	0.1	Km/Min
v_f	0.005	Km/Min
$[a_n, b_n]$	[0,480]	Min
T_a	5	/
β	0.2	/
d_i	70	Min
c_{sf}	{200,300,500}	/
c_{it}	0.5	/Min
c_{sv}	{1,1.5,2.5}	/Min
r_{sn}	{50,75,125}	/
r_{cross}	15	/
r_{cus}	15	/
r_{pt}	5	/Min
r_{nur}	5	/Min
c_d	0.2	/
c_w	0.5	/Min

Table 5

The set of parameters for algorithm.

Parameter	Value
Monte-Carlo simulation times M	100
Initial temperature T_0	1000
Final temperature T_f	0.1
Boltzmann's constant value K	0.9
Inter iteration times $maxInter$	300
Outer iteration times $maxOuter$	1000

Table 6

Number of patients and caregivers in test instances.

Size	ID	Number of caregivers				Dummy skill capability		
		$\delta_n = 1$	$\delta_n = 2$	$\delta_n = 3$	Total	δ	q_{si}	Q_{sn}
50	50_n	4	2	2	4	$s = 1$	10	50
100	100_n	8	4	4	8	$s = 2$	60	300
200	200_n	16	8	8	16	$s = 3$	310	1550

in Solomon's VRTPW benchmark: R instances, C instances, and RC instances. Since communities in China are dense, we choose C instances which include clustered locations for customers. The two-dimensional locations and deterministic service times of each customer are generated by referring to this benchmark. The sizes of 50, 100, and 200 are considered for the number of customers, and corresponding numbers of caregivers are 8, 16, and 32. The following modifications from the Solomon's benchmark are considered to adapt to our problem:

- (1) In this study, the coordinates of patients are three-dimensional. The specific coordinates position is generated based on the actual positions of the real-world case as mentioned in Section 5.1. Specifically, we use the relevant services provided by Baidu Maps to obtain community coordinates. Subsequently, we code on Matlab to randomly generate two-dimensional coordinates of patients within each community. Finally, the third-dimensional coordinates of patients are randomly generated according to the actual floor height.

Table 7Results for the test instances when $\rho = 50\%$.

Size: 50			Size: 100			Size: 200		
ID	SA	GA	ID	SA	GA	ID	SA	GA
50_1	1796.15	2030.08	100_1	2430.37	2589.92	200_1	4262.32	4664.67
50_2	1358.1	1569.65	100_2	2051.07	2237.17	200_2	4548.89	4842.22
50_3	1821.84	1855.21	100_3	2454.83	2670.7	200_3	4177.41	4522.76
50_4	1797.84	1859.39	100_4	2498.75	2823.95	200_4	4471.04	4917.99
50_5	2008.04	2149.99	100_5	2707.92	2842.28	200_5	4875.72	5302.97
50_6	2171.94	2226.94	100_6	2853.26	2971.04	200_6	4692.77	4983.31
50_7	1803.12	1993.73	100_7	2206.64	2375.59	200_7	4474.05	5013.54
50_8	1920.87	1937.2	100_8	2333.91	2387.94	200_8	4301.46	4619.57
50_9	1874.15	1935.65	100_9	2781.95	2931.26	200_9	4191.09	4569.68
50_10	2230.9	2370.39	100_10	3023.21	3271.6	200_10	4220.43	4709.54

Table 8Results for the test instances when $\rho = 70\%$.

Size: 50			Size: 100			Size: 200		
ID	SA	GA	ID	SA	GA	ID	SA	GA
50_1	1895.21	2027.64	100_1	2672.9	2873.15	200_1	4668.7	5041.86
50_2	1463.1	1638.13	100_2	2223.63	2426.61	200_2	4792.37	5193.32
50_3	1925.57	1940.19	100_3	2616.46	2721.85	200_3	4373.74	4720.44
50_4	1865.09	1904.77	100_4	2683.75	2914.72	200_4	4681.16	5154.27
50_5	2082.23	2157.74	100_5	2945.18	3142.93	200_5	5325.15	5641.33
50_6	2243.45	2282.37	100_6	3116.88	3236.76	200_6	4903.86	5219.66
50_7	1905.36	2039.95	100_7	2379.76	2593.55	200_7	4900.08	5391.99
50_8	2036.73	2092.61	100_8	2535.99	2602.84	200_8	4411.3	4801.34
50_9	1976.58	2034.04	100_9	3045.28	3214.34	200_9	4419.08	4785.18
50_10	2322.13	2491.18	100_10	3291.34	3516.59	200_10	4537.38	4898.45

Table 9

Experimental results for the small-sized instance.

Scheduling method	Cross-regional scheduling			Independent scheduling		
	A	B	Total	A	B	Total
Care center						
Total cost	398.41	302.91	701.32	394.1	469.04	863.14
Fixed cost	1500	700	2200	1500	700	2200
Travel cost	174.48	116.51	290.99	140.1	63.05	203.15
Overtime cost	0	7.58	7.58	0	0	0
Service revenue	1625	775	2400	1625	650	2275
Cross-regional revenue	-60	60	0	/	/	/
Penalty for unvisited patients	75	165	240	120	135	255
Penalty for violating TW	30.25	0	30.25	0	0	0
Penalty for overtime working	0	25.25	25.25	0	0	0
Penalty for overqualification	34.4	15.2	49.6	106.8	12.4	119.2
Penalty for waiting times	149.28	108.38	257.66	152.2	208.59	360.79

Table 10

Experimental results for the middle-sized instance.

Scheduling method	Cross-regional scheduling			Independent scheduling		
	A	B	Total	A	B	Total
Care center						
Total cost	360.54	385.81	746.35	667.49	378.39	1045.88
Fixed cost	3000	1700	4700	3000	1700	4700
Travel cost	339.74	205.82	545.56	289.93	150.12	440.05
Overtime cost	66.7	0	66.7	3.24	0	3.24
Service revenue	3650	1950	5600	3325	1950	5275
Cross-regional revenue	60	-60	0	/	/	/
Penalty for unvisited patients	225	120	345	195	195	390
Penalty for violating TW	3.65	3.9	7.55	0	0	0
Penalty for overtime working	105.75	0	105.75	10.8	0	10.8
Penalty for overqualification	99.6	99.2	198.8	236.8	38.8	275.6
Penalty for waiting times	230.1	146.89	376.99	256.72	244.47	501.19

(2) There are two regions in our study, so the proportion of patients and caregivers in the two regions must be considered. The proportion of patients in the two regions is 3 to 2, which is set according to the number of residential buildings. The actual operation situation of both care centers is unknown to us;

however, it can be known that the care center A is a little bigger in scale because it belongs to a chain of medical institutions. Thus, the proportion of caregivers in the two regions is 5 to 3. There is no other HHC center in the area, so the manpower of the care center A is more sufficient considering the caregiver–patient

Table 11
Experimental results for the large-sized instance.

Scheduling method	Cross-regional scheduling			Independent scheduling		
	A	B	Total	A	B	Total
Total cost	697.9	761.99	1459.89	1023.05	940.6	1963.65
Fixed cost	6000	3400	9400	6000	3400	9400
Travel cost	647.48	464.23	1111.71	556.39	314.74	871.13
Overtime cost	151.27	41.37	192.63	8.94	16.14	25.07
Service revenue	7250	3975	11 225	6900	3850	10 750
Cross-regional revenue	15	−15	0	/	/	/
Penalty for unvisited patients	345	345	690	390	390	780
Penalty for violating TW	0	0	0	107	0	107
Penalty for overtime working	119.1	81.25	200.35	42.05	42.8	84.85
Penalty for overqualification	155.2	80.4	235.6	301.6	77.6	379.2
Penalty for waiting times	544.86	309.74	854.6	517.07	549.33	1066.4

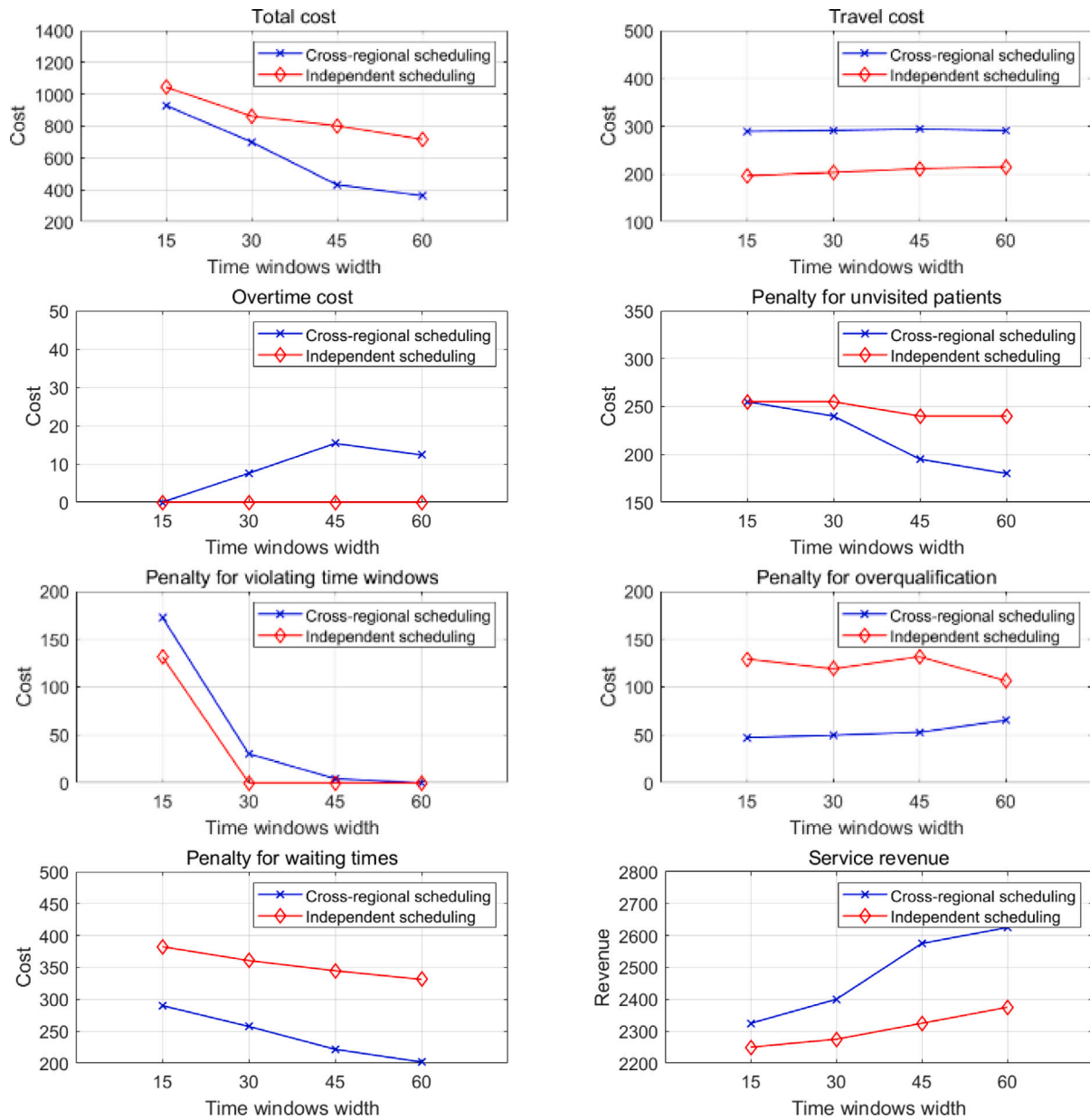


Fig. 7. Impact of time windows width of small-sized instance.

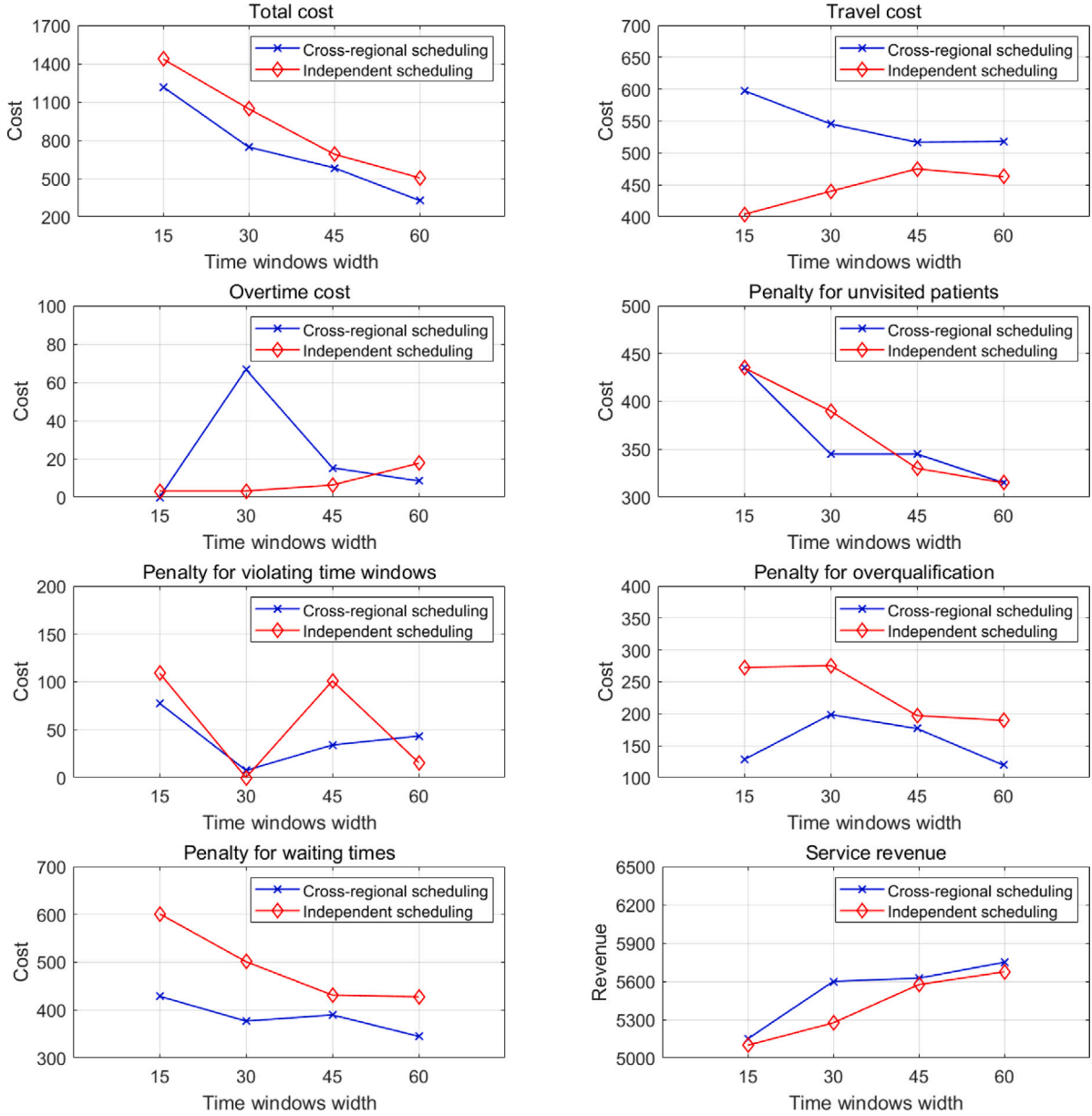


Fig. 8. Impact of time windows width of middle-sized instance.

ratio. The number of patients and caregivers in each instance can be seen in Table 3.

- (3) Three levels of skill qualification are considered based on the service portfolio provided by the care center A. For each task required by patients, dummy skill demands will be generated to complete the skill matching constraint. Similarly, each caregiver has his or her dummy skill capability.
- (4) In reality, time windows of patients will not be too wide, so we set it to 30 min. To be more realistic, patients only start the time window at the beginning or the half of an hour.

In this paper, instances are named as CR-size to be separated from the Solomon's benchmark. The main parameters for objective function and algorithm are shown in Tables 4 and 5, respectively. The parameter value is determined as follows: v_e and v_f are travel speeds, which refer to the actual electric bicycle and pedestrian speeds. The normal speed

of electric bicycles in urban areas is 15 km/h. Since the caregivers may need to load and unload equipment, wait for traffic lights, etc., a more conservative value (6 km/h) is set to ensure that caregivers are not easy to be late. d_i , T_a are determined by interviewing the operators of home health care institutions. c_{sf} refers to the information of the recruitment website. r_{sn} is based on the information disclosed on the website of the care center A. c_{cus} , c_{pt} , c_{nur} , c_d and c_w need to be formulated by decision makers according to the company's operating policies. For example, we set c_{pt} to 5 and c_w to 0.5 in this paper because do not want patient dissatisfaction due to caregivers arriving late. As a sacrifice, our final plan may result in longer waiting times for caregivers. r_{cross} refers to the coefficient of additional benefits for providing cross-regional services, which is determined according to the actual market conditions. We interviewed three home health care facility operators and determined the final value based on their recommendations.

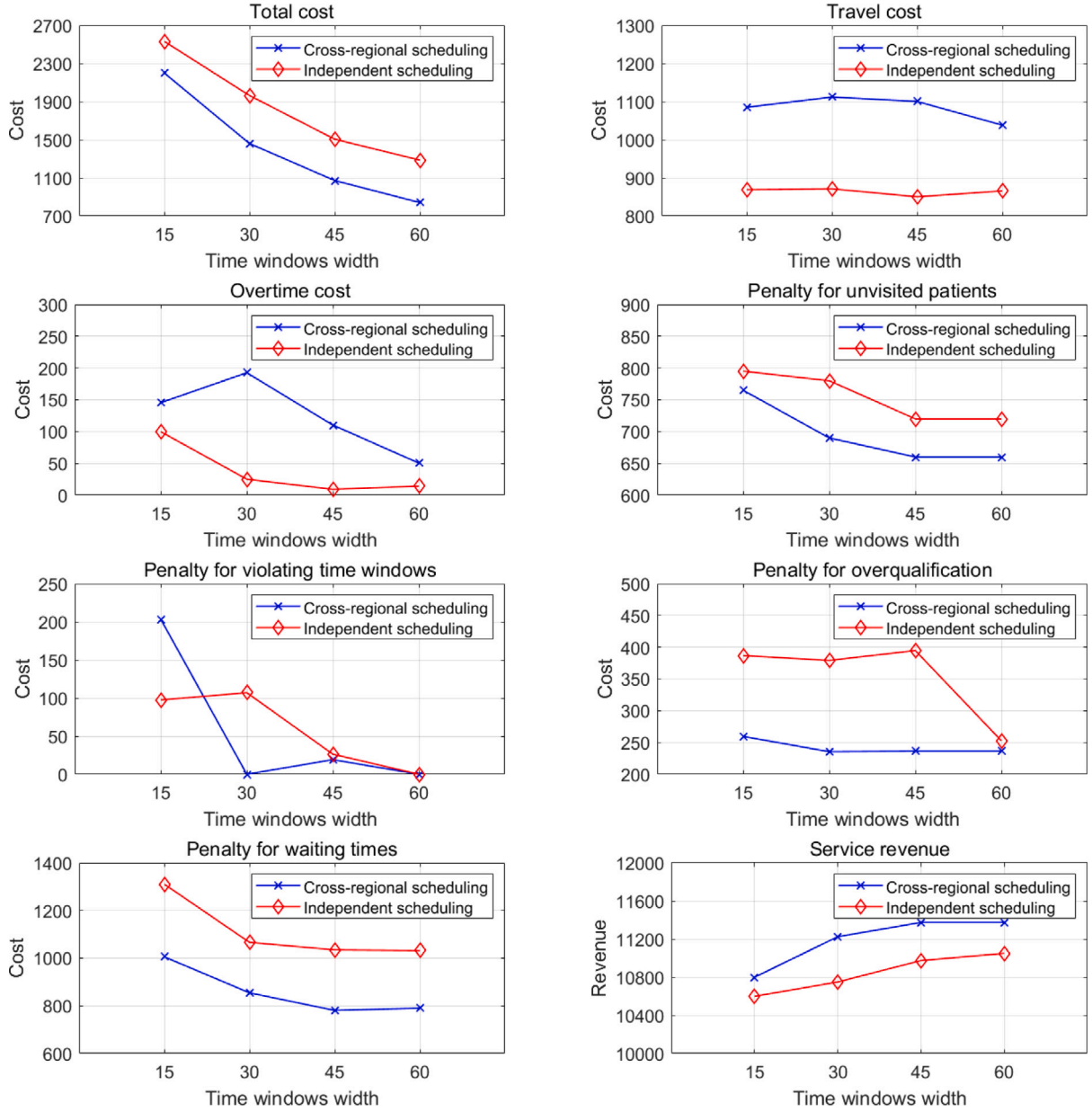


Fig. 9. Impact of time windows width of large-sized instance.

The proposed algorithm is executed on a 50 GHz Intel®Core™i5-7300HQ CPU. The computational time is limited within 1 h, 2 h, and 4 h for the small, middle, and large instances. For each instance, we run it ten times to get the best solution as the final result. The independent scheduling of each region will be operated ten times to get the best one in the same way. In the next part, the results of cross-regional and independent scheduling will be emphatically compared.

5.3. Experimental results

5.3.1. Results of test experiments

Since there is no similar model in the existed literature, we generate some test instances to verify the performance of the hybrid Simulated Annealing algorithm. The proposed model is reduced to a unidirectional cross-regional scheduling problem. That is, only the care center

A provides cross-regional services to the region B. The number of caregivers and dummy skill capabilities in test instances can be seen in Table 6. As for the number of patients, suppose that ρ is the proportion of patients in the region A. The test experiments are conducted setting that $\rho = 50\%$ and $\rho = 70\%$. Genetic Algorithm (GA) is used to get the benchmark result. The computational time of GA is set as the same as SA to ensure the comparability of results. The results of test experiments can be obtained from Tables 7 and 8. In all the test instances, the proposed hybrid SA gets better solutions compared with the contrast algorithm in the same computing time.

5.3.2. Comparing analysis of cross-regional and independent scheduling

Subsequently, the empirical analysis is conducted based on the settings in Section 5.2. Each instance is scheduled by both cross-regional method and independent method. The results of small, middle,

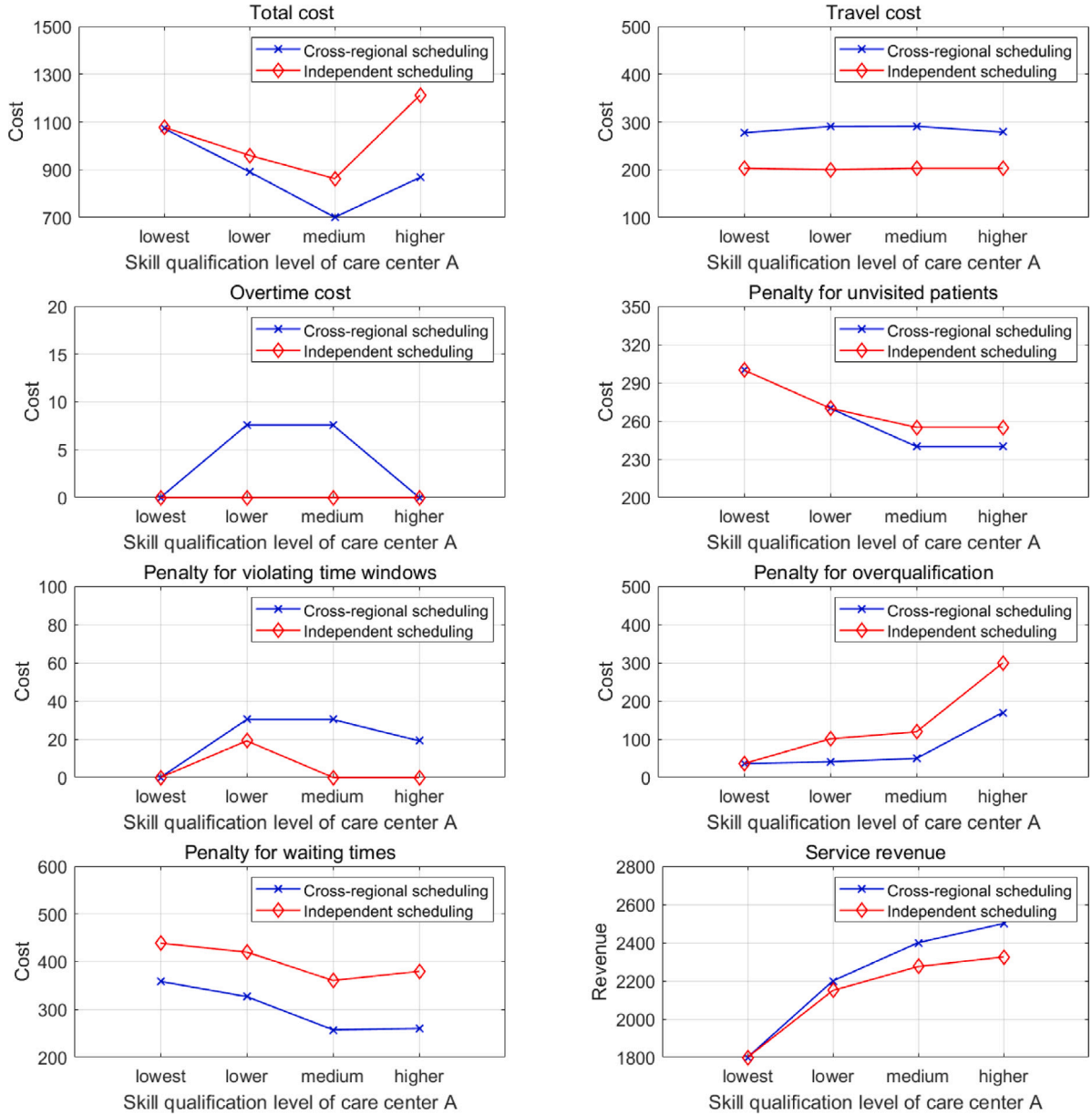


Fig. 10. Impact of skill qualification structure of small-sized instance.

and large instances can be obtained from Tables 9, 10, and 11. As can be seen in the Tables, cross-regional scheduling reduces the total costs compared with independent scheduling. The characteristics of cross-regional scheduling are concluded as follows:

1. For care centers, cross-regional scheduling can increase service revenues. More patients will be served in the one-day planning horizon. It is beneficial for care centers to adopt cross-regional scheduling to make full use of personnel resources. Though travel costs and overtime costs increase, it can be compensated by the increase of service revenues.
2. For patients, more tasks can be covered, which improves the patients' overall welfare. As is illustrated in Section 1, home medical care resources are in short supply in China. Cross-scheduling can eliminate the gap between supply and demands to some extent. In large-sized instance, cross-regional scheduling also reduces the penalty for violating patients' time windows.
3. For caregivers, the waiting times and overqualification decrease in the cross-regional situation; however, caregivers will work longer time than in the independent situation.

5.3.3. Impact of time windows width

To investigate the effect of time windows width, the patients' time window widths are set as 15, 30, 45, and 60 min to compare the cost differences between cross-regional and independent scheduling. The results are presented in Figs. 7, 8, and 9. As can be seen in the Figures, the total cost decreases with time windows width increases. In all cases, the results of cross-regional scheduling are better than independent scheduling. The results reveal that cross-regional scheduling is beneficial to reducing the total cost whatever the time windows width policy the policymakers decide to adopt. The conclusions in Section 5.3.2 are relatively robust considering the time windows width.

5.3.4. Impact of caregivers' skill qualification structure

The relative skill qualification structure of two centers may affect the results of cross-regional scheduling. To investigate the specific effect, the caregivers' overall skill qualification of the care center A is set as lowest, lower, medium, and higher to compare the cost differences between cross-regional and independent scheduling. The results are presented in Figs. 10, 11, and 12. When the skill qualification

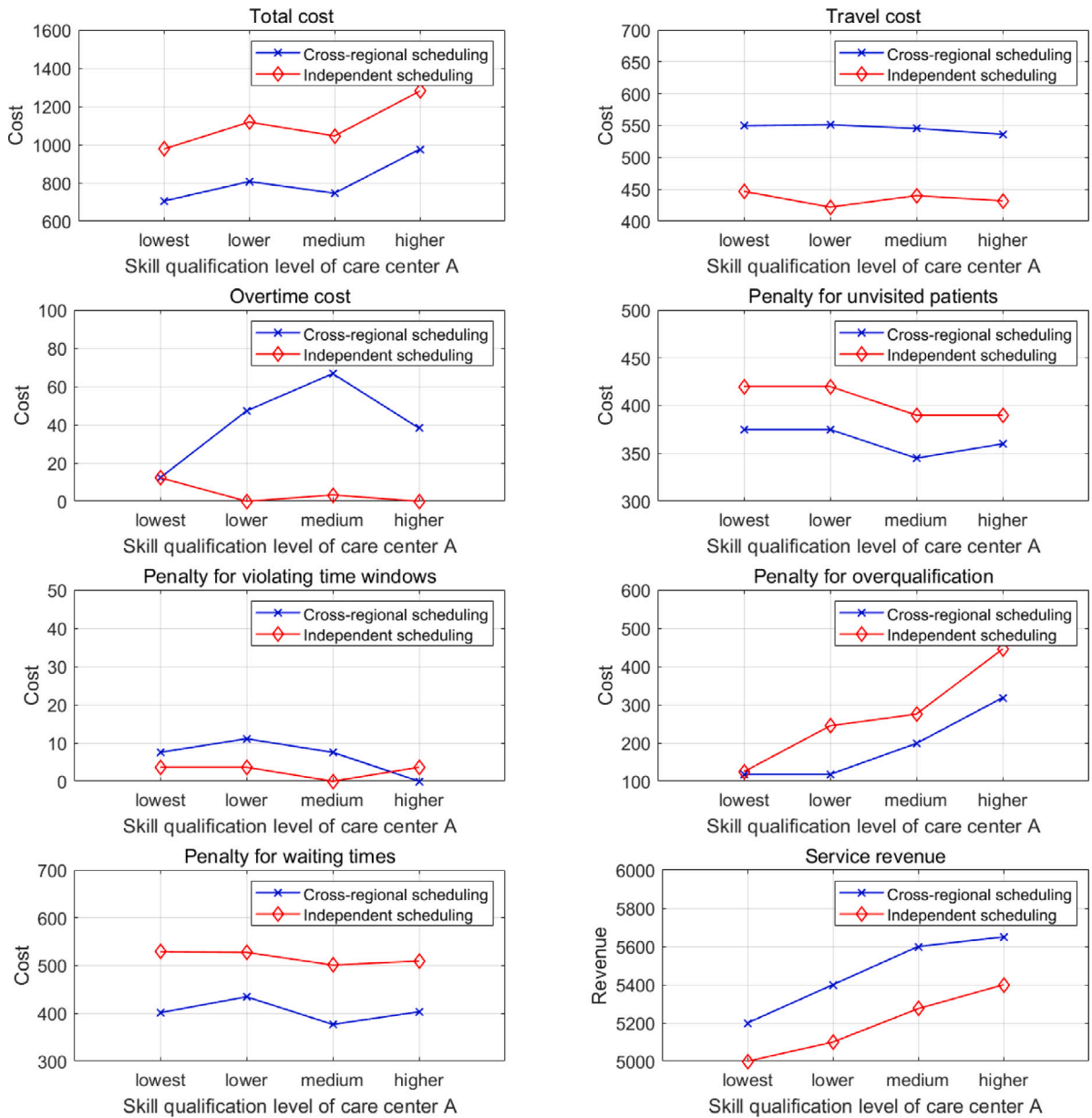


Fig. 11. Impact of skill qualification structure of middle-sized instance.

level of care center A is medium, both cross-regional and independent scheduling methods have the best results. When the skill qualification level of care center A is higher, the total cost will increase due to the increase of overqualification penalty. Thus, the conclusions in Section 5.3.2 are also robust considering the skill qualification structure of care centers.

6. Conclusion

This paper proposes a cross-regional scheduling and routing problem in HHC with service times in a one-day horizon. Some real-life assumptions are added into the mixed integer linear programming model, such as patient time windows, working time regulations, task importance, and skill matching. To solve the problem, this paper develops a hybrid Simulated Annealing algorithm framework to solve the proposed model. We propose an initial solution generation method and a neighborhood search strategy, and use the flexible memory structure of the TS algorithm to improve the algorithm. For the uncertainty of service time, we use Monte Carlo simulation method to evaluate the

cost of the uncertain problem. Then, this study compares the results of cross-regional scheduling and independent scheduling in the two regions based on the real cases of two care centers in Minhang District, Shanghai. From the experiments, we conclude that cross-regional scheduling is beneficial to reduce the total cost of two care centers. The sensitivity analysis verified the robustness of the above conclusions.

Our analysis reveals five important management insights. First, we found that the cooperation of HHC enterprises in cross-regional scheduling is conducive to reducing the total scheduling cost and improving the overall resource utilization efficiency. Unified standards are conducive to reducing the cost of negotiation and disputes among enterprises. Therefore, the government can create a cooperative environment for HHC enterprises by formulating various norms. Second, in the case experiment set up in this study, when the size of the care center is smaller, the benefit level of care center B with the lower skill level of caregiver is higher, while the total cost of care center A increases. In an operational environment, care center A would have no incentive to collaborate with care center B for cross-regional scheduling. Therefore, enterprises need to conduct sufficient market research

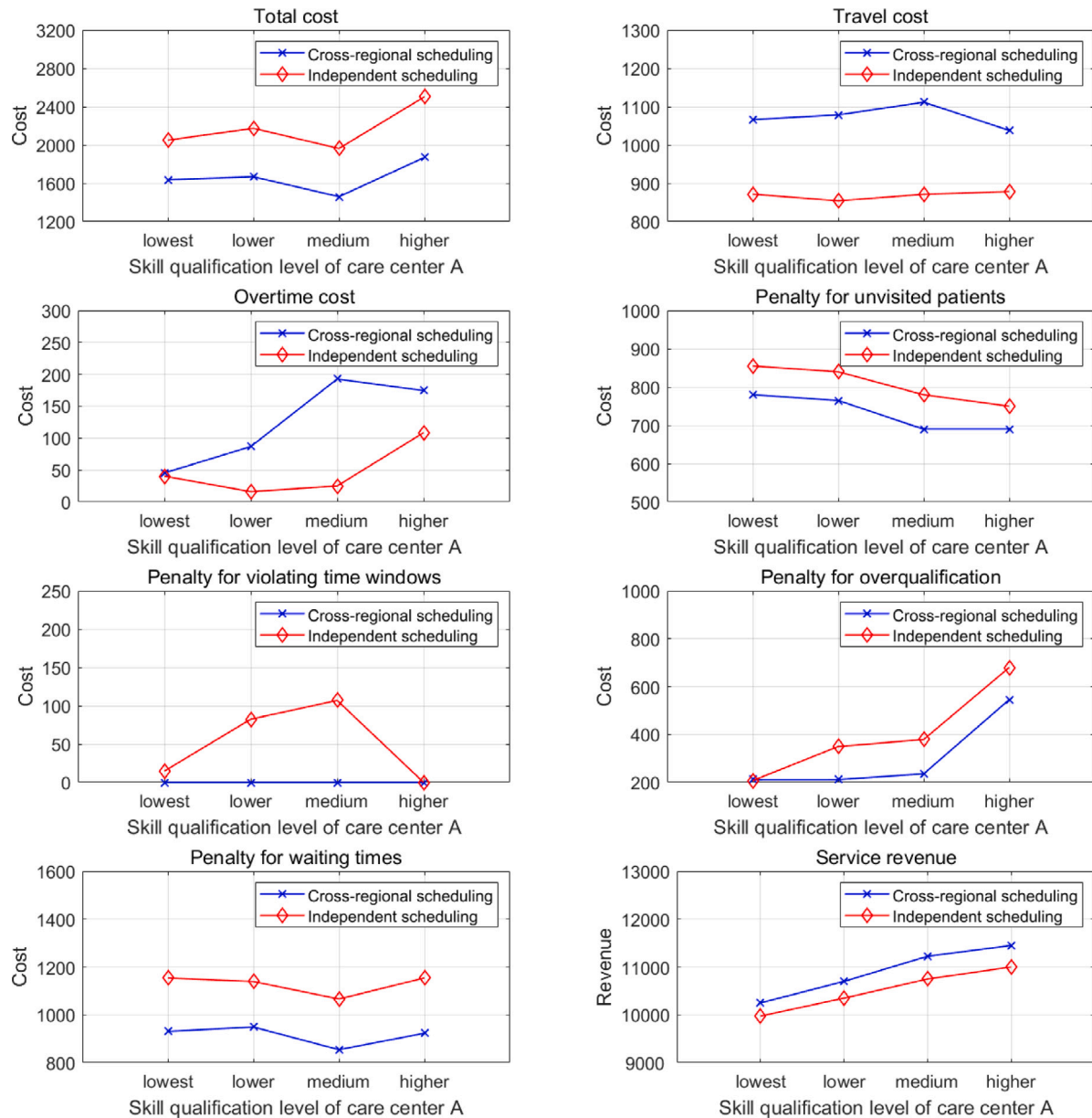


Fig. 12. Impact of skill qualification structure of large-sized instance.

to understand their overall positioning in the market when choosing partners. Third, in cross-regional scheduling, enterprises should fully evaluate their own resource usage and whether potential cooperative enterprises complement their own resources, so as to achieve a win-win situation in cooperation. Fourth, the number and skill allocation of nursing staff directly affects the number and types of services that can be provided. In order to provide patients with more variety and high-quality service portfolio solutions, enterprises need to establish a caregiver team matching the needs of patients. Finally, the enterprise should improve the patients' expectation of service quality, such as not giving patients too strict commitments without affecting their health, and encouraging patients to adopt a looser time window policy.

We highlight some limitations of our study that should be considered in future work. First, we can further explore various factors that affect cross-regional scheduling, and analyze the boundary conditions for cross-regional scheduling and the pricing mechanism for cross-regional collaborative scheduling through more experiments. Second, this study only considers the cross-regional scheduling situation of two care centers, so the collaborative scheduling problem of multi-region

and multi-center can be further studied in the future work. Finally, the cost objectives and parameters of this study are set with reference to existing research, such as factors affecting the satisfaction of patients and caregivers. In future work, relevant empirical studies or machine learning methods can be used to accurately evaluate or estimate various cost parameters in the scheduling problem of HHC caregivers.

CRedit authorship contribution statement

Gang Du: Conceptualization, Methodology, Investigation, Visualization, Formal analysis, Funding acquisition, Supervision, Writing – review & editing. **Jingjing Zhang:** Data curation, Software, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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All authors read and approved the manuscript.

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